AI-based Project for cybersecurity (AI4Cyber) Assignment Description

Assignment Description

This assignment aims to guide students through the development of a complete machine learning project, focusing on real-world cybersecurity applications that involve both regression and classification tasks. The assignment challenges students to apply their technical skills to practical cybersecurity scenarios. Students are expected to acquire the following skills from the assignment:

- **Data Collection and Processing:** Gathering and preprocessing data from open-source websites, ensuring that the data is clean, normalized, and ready for analysis.
- **Model Selection and Evaluation:** Selecting appropriate machine learning models based on the problem context, and evaluating their performance using relevant metrics such as accuracy, precision, recall, and F1 score.
- **Technical Implementation:** Gaining hands-on experience with Python and key libraries like pandas, scikit-learn, and matplotlib for data handling, modeling, and visualization.
- **Report Writing and Presentation:** Writing a structured report that clearly communicates the project objectives, methodologies, findings, and conclusions, supported by appropriate visualizations and references.

Students have already chosen the following cybersecurity-related topic in assignment one:

1. **Misinformation Detection on Social Media:** Detect misinformation on social media platforms. Students will collect real-world social media data, extract useful features (like text content and user behaviour), and train models to identify false or misleading posts.
2. **Spam or Malware Detection:** Detect spam messages or potential malware threats. Students will work with datasets containing emails, messages, or files, extract relevant features, and build models to classify harmful content.
3. **Detecting Software Vulnerabilities:** Detect security vulnerabilities in software code. Students will analyse source code to identify patterns that may lead to bugs or exploits, and train models to automatically flag risky code.
4. **Network Traffic Classification for Anomaly Detection:** Classify network traffic and detect unusual or potentially malicious activity. Students will analyse network data, extract features such as packet size and timing, and train models to distinguish normal behaviour from anomalies.
5. **Cybersecurity threat detection in banking:** This covers a range of cybersecurity problems in the banking sector. Students are expected to do some research to choose a

suitable cybersecurity problem (e.g. fraud detection, identify theft, etc.) to solve using AI engineering.

Key Information

- **Weight:** 40% of your total marks for the unit
- **Due Date:** Due in Week 8
- **Submission Requirements:**
    - **A report:** Size up to 3000 words; Format PDF
    - **A zip file containing:**
        - Datasets that you have already processed (JSON/CSV and so on)
        - Readme file (pdf/md): telling us how to run your code
        - Your source code (including the final models you used)
    - **Meeting Minutes:** Format PDF
    - **Contribution Form:** Format PDF
- **Late Penalties:** 10% deduction of the available mark per calendar day or part thereof for up to one week. Submissions that are late more than 7 calendar days after the due date will receive a mark of zero (0) and no assessment feedback will be provided.

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1. Project Description

The project description provides an overview of the background and essential requirements for the cybersecurity-related projects. This project involves developing a comprehensive machine-learning solution that integrates project management, design elements, and technical implementation to address real-world cybersecurity challenges. Students will work in teams to complete the project in three phases: creating a detailed project management plan, implementing a machine learning model, and developing a dynamic website to showcase their results.

The Machine Learning Web Application aims to deliver an interactive platform for users to engage with machine learning models and visualize data insights. The primary goal is to demonstrate practical machine learning applications in real-world cybersecurity-related scenarios, enhancing user interaction and understanding of the underlying models.

Core Functional Requirements for the whole project:

- Select a project topic and find relevant datasets suitable for the chosen topic.
- Investigate and analyse the topic, using machine learning techniques for prediction, attribution, or classification to gain a better understanding and response to the cybersecurity-related topic.

- Develop a website application that allows users to interact with trained models and provides corresponding responses based on user inputs.
- Include data visualization features to help users understand the dataset.

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2. Submission

You must submit your assignment via the assignment submission link (i.e., “Assignment 2 Submission”) on the Canvas site by the deadline.

- No hard copy submission is required for this assignment.
- You are required to submit your assignment named with your group name. For example, if your group name is “Session01-Group 1”, you would submit the files named “session-xx-group-1-AssignmentName.pdf”.
- Do not include any unnecessary files in this folder.
- Note that marks will be deducted if this requirement is not strictly complied.
- No submission is accepted via email.

****Only one student per group is required to submit all the assignment documents on behalf of the group.** **

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3. Overview of the Tasks

- Use the topic you selected for the project.
- Clearly define the intended users of your project. The user might be the elderly, young children, or whoever you like, although the choice should make sense for the data and topic of choice.
- **Data Collection:** Use open-source websites to collect relevant data. Ensure the data is appropriate for the selected topic and intended users.
    - *Note:* Using only one simple dataset will earn you a basic score. To achieve a high distinction (HD) grade, you need to find and use additional suitable datasets.
- **Data Processing:** Clean and preprocess the collected data. This may include handling missing values, normalizing data, and feature extraction to prepare the data for analysis.
- **Data Analysis:** Analyze the preprocessed data to uncover patterns and insights. Utilize statistical methods and visualizations to better understand the data and inform the selection of the machine learning model.
- **Model Selection:** Choose an appropriate machine learning model based on the data and the problem you are addressing. Consider models such as regression, classification, clustering, etc.

    - *Note:* Students are required to use at least two types of machine learning methods (classification, clustering, regression) out of the three. Students may use two of the same type, but they must provide strong justifications. To earn additional HD points, conduct extra machine learning experiments beyond the existing two algorithms.
- **Model Evaluation:** Evaluate the chosen model's performance using relevant metrics (accuracy, precision, recall, F1 score, etc.).
- **Implementation in Python:** Execute all tasks using Python, leveraging libraries such as pandas, scikit-learn, and matplotlib.
- **Report:** Compare the evaluation metrics and present the findings in a comprehensive report, including visualizations, process explanation, and results interpretation.

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4. Deliverables

Your submission should contain the following files:

1. Report

Write a report of up to 3000 words (excluding cover page, table of contents, bibliography, and appendix) that consists of the following sections:

- **Project title:** Title of your machine learning project.
- **Your group identities:** Your group name and ID, student names and IDs, tutor's name.
- **Introduction:** A precise and succinct description of what motivations you wanted your machine learning project to solve, and who the intended user is.
- **Problem Framing:** Outline the specific challenge you aim to address, limitations of existing solutions, and why an ML approach is suitable.
- **Data Collection:** Detail the sources and methods used to gather your dataset. Mention any challenges encountered and how they were addressed.
- **Data Processing:** Outline steps taken to clean and preprocess the collected data (handling missing values, normalization, feature engineering, joining datasets).
- **Machine Learning Model Selection:** Describe the criteria used to select models, algorithms considered, and rationale behind your choices.
- **Implementation:**
    - *Technical Implementation:* High-level description of implementation, libraries used, and reasons for final decisions.
    - *Implementation Evaluation:* Evaluate effectiveness of implementation by comparing results against objectives. Discuss model performance and highlight unexpected outcomes.
- **Conclusion:** Summarize key findings and outcomes, discuss significant insights, and potential areas for future work.

- **Bibliography:** Appropriate references in Harvard style.
- **Appendix:** Additional files.

2. A zip file

- **Datasets:** The dataset must only include the data used in your final version machine model.
- **Readme file (pdf/md):** Explain how to configure your project environment using conda commands, perform data processing, train your model, and use your model for prediction.
- **Source code:** Any code related to your assignment 2.

3. Meeting Minutes

Students are required to hold at least one meeting each week since the team's inception and submit all meeting minutes.

4. Contribution Form

A form including details of the contribution and personal information. All team members must discuss and sign together within the group before submitting.

Important Notes:

- Please be careful to ensure you do not publicly post anything which includes your reasoning or logic to Canvas discussions to avoid plagiarism/collusion penalties.
- You must **NOT** use generative artificial intelligence (AI) to generate any materials or content related to the assessment task.

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5. Marking Criteria

You must acknowledge all statements and information taken from other sources (Harvard style).

Rubric (COS30049 Assignment 2)

Criteria	**Ratings**	**Pts**
Report: 1) Explanation of Data Collection and Processing Operations	**5 Pts:** Comprehensive and clear. **3 Pts:** Adequate with minor gaps.	5 pts

Criteria	Ratings	Pts
	1 Pts: Limited with significant gaps. **0 Pts:** Missing or inadequate.	
Report: 2) Explanation of Data Analysis	**2 to >1.0 Pts:** Detailed and clear. **1 to >0.0 Pts:** Basic explanation. **0 Pts:** Missing or insufficient.	2 pts
Report: 3) Justification of Model Selection	**2 Pts:** Thorough justification. **1 Pts:** Limited justification. **0 Pts:** Missing or poorly articulated.	2 pts
Report: 4) Quality of Writing and Logical Structure	**1 to >0.0 Pts:** Exceptionally well-written, clear structure, no errors. **0 Pts:** Poorly written, lacks coherence.	1 pts
Report: 5) Academic Referencing and Citations	**1 Pts:** All sources correctly cited in Harvard style. **0 Pts:** Significant errors or missing references.	1 pts
Report: 6) Completeness	**1 Pts:** All required sections included. **0 Pts:** Several sections missing or incomplete.	1 pts
Source code: 1) Core Functionalities	**10 Pts:** Data thoroughly processed; models well-selected and evaluated. **8 Pts:** Minor gaps in documentation/evaluation.	10 pts

Criteria	Ratings	Pts
	4 Pts: Significant gaps in documentation/justifications.	
Source code: 2) Datasets	**4 to >3.0 Pts:** Complete, highly relevant, clear format. **3 to >1.0 Pts:** Includes irrelevant data or formatting errors. **1 to >0 Pts:** Incomplete, irrelevant, or incorrect format.	4 pts
Source code: 3) readme file	**2 to >1.0 Pts:** Comprehensive instructions for env setup, training, and prediction. **1 to >0 Pts:** Incomplete, confusing, or poor presentation.	2 pts
Source code: 4) Comments	**2 to >1.0 Pts:** Thoroughly commented. **1 to >0 Pts:** Minimal or vague comments.	2 pts
Source code: 5) Code Structure	**2 to >1.0 Pts:** Clean, well-organized, modular. **1 to >0.0 Pts:** Generally organized but some issues. **0 Pts:** Poorly structured.	2 pts
Additional datasets	**4 to >2.0 Pts:** Utilize large number of additional datasets. **2 to >0 Pts:** Moderate or small amount.	4 pts

Criteria	Ratings	Pts
Additional ML implementations	**4 to >2.0 Pts:** Appropriate methods, well-tuned, excellent performance. **2 to >0 Pts:** Methods not entirely appropriate, flaws in implementation.	4 pts